# Master of Applied Technologies

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| **Course No** | COMP8831 |
| **Course Name** | Machine Learning |
| **Level** | 8 |
| **Credits** | 15 |

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|---|---|
| **Student Name** | ______________________ |
| **Student ID** | ______________________ |
| **Assessment Type** | Final Project: Report and Presentation |
| **Weighting** | 60% |
| **Due Date and Time** | Friday, 12/06/2026 23:59 |
| **Total Marks** | 100 (Report 50 + Code & Presentation 50) |

**Student Declaration**

I confirm that:

- This is an original assessment and is entirely my own work.
- The work I am submitting for this assessment is free of plagiarism. I have read and understood the Academic Integrity Procedure. I have also read and understood the Student Disciplinary Statue.
- Where I have used ideas, tables, diagrams etc of other writers, I have acknowledged the source in every case.

**Students Signature:** ______________________

**Date:** ______________________

# 1 Assignment Aims

- To implement a machine learning project that integrates and applies the skills and knowledge gained through the course.
- To identify, apply, and critically evaluate a collection of machine learning algorithms against alternative methods.
- To achieve greater awareness of the latest artificial intelligence tools and techniques for prediction, recognition, optimisation, and decision making.
- To understand the principles, advantages, limitations, and possible applications of machine learning algorithms in different areas.

# 2 Assignment Task

This final project is the implementation of your **approved proposal**. You will take your project through the full cycle, from research question to evidence-based recommendation: execute the methodology you proposed, run experiments, critically evaluate your results, and defend your conclusions in a presentation. The final project builds directly on the plan you laid out in the Project Proposal Assignment. Expand your methodology into working code, report your experimental results, and reflect on what worked and what did not.

The final project weighs **60 percent** of your final mark and is assessed through two parts: a written **Report** (50 marks) and a combined **Code & Presentation** deliverable (50 marks). **Each group submits only one version of each deliverable**, with the names of all group members included on the first page of the report, in the code README, and on the title slide of the presentation.

**Important Requirements**

- **Build on your approved proposal.** Your final project must implement the plan approved by the lecturer. Significant changes in scope, dataset, or methods require prior lecturer approval.
- **Model Requirements:** Your primary solution must use modern, state-of-the-art machine-learning techniques (e.g., neural networks, advanced ensemble models, deep learning architectures, transformer variants, graph neural networks, diffusion models).
- **Baseline Comparisons:** Classical methods (logistic or linear regression, SVMs, random forests, decision trees) are welcome only as baseline benchmarks. They must not be your primary method.
- **Group Size and Methods:** Projects may be completed individually or in groups of up to four. A group of **N** members must implement and compare at least **N** distinct learning methods. Individual projects must still compare multiple models.

# 3 Part A: Project Report (50 Marks)

Your report should be between **4000 and 8000 words**, **excluding** references and appendices. A suggested structure is given below. The word counts and the detailed contents of each subsection are **recommendations, not fixed requirements**; use them as a guide to plan your own report. The sections themselves should all appear; how you scope the content inside each is your decision.

## 3.1 Title Page

Include the project title, full name(s) of group member(s), student ID(s), course code, and date.

### 3.2 Table of Contents

List all sections and subsections with page numbers.

### 3.3 Abstract *(200–300 words)*

A concise summary of the project. Cover:

- The problem addressed.
- The machine learning approach(es) implemented.
- The key quantitative results.
- The main conclusion and recommended model.

### 3.4 Introduction & Literature Review *(800–1200 words)*

Expand on the Problem Statement and Background sections of your proposal.

- Describe the problem and its real-world significance.
- Critically review at least **10 quality sources** (peer-reviewed papers, conference proceedings, books, or reputable technical resources). Do not just summarise; compare approaches, identify gaps, and explain how your work relates to existing work.
- State your research questions or hypotheses.

### 3.5 Dataset & Exploratory Data Analysis *(600–1000 words)*

Document the data you used and how you prepared it:

- **Source and licensing:** cite the dataset source; include a direct link or reference so it can be accessed by others.
- **Data dictionary:** list all features and their types (categorical, numerical, text, image, etc.).
- **Exploratory analysis:** include visualisations of feature distributions, correlations, missing-data patterns, and class imbalance (where applicable).
- **Preprocessing pipeline:** feature engineering, scaling or normalisation, encoding, and any dimensionality reduction.
- **Train / validation / test split:** state proportions and justify the split strategy (random, stratified, temporal, grouped, etc.).

### 3.6 Methodology *(800–1200 words)*

Describe how you implemented and trained your models:

**Methods Implemented**
Describe each ML method you implemented (at least N for an N-member group). For methods not covered in class, provide the technical detail a reader needs to understand your choices.

**Block Diagram / Flowchart**
Include a diagram that maps the full pipeline: *Data* $\rightarrow$ *Preprocessing* $\rightarrow$ *Split* $\rightarrow$ *Model Training* $\rightarrow$ *Hyperparameter Tuning* $\rightarrow$ *Evaluation*.

**Hyperparameter Strategy**
Describe how you selected hyperparameters (grid search, random search, Bayesian optimisation, manual tuning) and the ranges explored.

**Experimental Setup**
Tools, libraries, frameworks, hardware (CPU/GPU), random seeds used for reproducibility.

### 3.7 Results & Discussion *(800–1200 words)*

Report and interpret your findings:

- **Quantitative metrics:** use metrics appropriate to the task: accuracy, precision, recall, F1 for classification; MAE, RMSE, $R^2$ for regression; task-specific metrics for other problems.
- **Diagnostics:** include learning curves, and confusion matrices or residual plots, to characterise model behaviour.
- **Model comparison:** a side-by-side table of all models on the same metrics, with a clear discussion of which method performs best and why.
- **Error analysis and bias/variance discussion:** where do the models fail? Which errors matter most? How does performance change with model complexity or training size?
- **Limitations:** what are the limits of your study (data, compute, scope)?

### 3.8 Conclusion & Future Work *(200–400 words)*

Summarise the key findings, state the recommended model, and identify concrete directions for future improvement.

### 3.9 References

List all sources cited using **IEEE referencing style**. A minimum of **10 quality sources** is required. Examples of IEEE formatting:
`http://libguides.murdoch.edu.au/IEEE/sample`

## 4 Part B: Code & Presentation (50 Marks)

Part B covers both the working implementation and the oral presentation. Code and presentation together are weighted equally with the report.

### 4.1 Code Submission

Submit a single zipped archive containing the working implementation. Suggested contents:

- **Code files:** Jupyter notebooks or structured `.py` files. Code should be well commented and organised; notebooks should include markdown explanations alongside the code.
- **Reproducibility:** fix all random seeds; ensure the code runs end-to-end on a fresh environment.
- **Environment:** include a `requirements.txt` or `environment.yml` listing package versions.
- **Data access:** include a link to the dataset or instructions for obtaining it. Do not upload large datasets directly unless licensing permits.
- **README:** a short file explaining how to run the code and reproduce the reported results, and listing all group member names.

### 4.2 Presentation

Each group gives a **10-minute** in-class presentation followed by a **5-minute Q&A**. The presentation should be **brief, to the point, and fluent**. **All members of the group must contribute** to the delivery, with an equal share expected.

Structure your 10 minutes around the following flow:

**Introduction**

State the problem, explain why machine learning is a suitable approach for this problem, and clearly state your research question.

**Methodology**
Introduce the dataset and give a *brief* (non-technical) overview of the models you used. Focus on *why* these models fit the task rather than the internal technical details.

**Results & Discussion**
Present the outcomes of your experiments and include a **summary table comparing all models and their performance** on your chosen metrics.

**Chosen Model**
Finish by stating which model you selected as your final solution and justify why.

The Q&A is your opportunity to demonstrate depth of understanding. Every member should be able to discuss the full project, not only the portion they prepared.

## 5 Marking Criteria: Report (50 Marks)

| Criteria | Marks | What We Assess |
|---|---|---|
| Organisation & Writing | 10 | Academic tone, logical flow, formatting, grammar, TOC and numbering correctness |
| Literature Review & Background | 5 | Critical engagement with $\geq$10 quality sources, clear link to your methodology |
| Dataset & EDA | 5 | Dataset documentation, exploratory analysis, preprocessing, train/val/test split justification |
| Methodology | 15 | Modern methods implemented, flowchart clarity, hyperparameter strategy, experimental setup |
| Results & Discussion | 15 | Appropriate metrics, learning curves / confusion matrices, model comparison, error analysis, limitations |
| **Total** | **50** | |

## 6 Marking Criteria: Code & Presentation (50 Marks)

| **Criteria** | **Marks** | **What We Assess** |
|---|---|---|
| *Code (20 marks)* | | |
| Implementation | 8 | Working end-to-end pipeline, methods from the proposal correctly implemented |
| Code Quality & Documentation | 6 | Readable, well-organised code; notebooks with markdown / meaningful comments |
| Reproducibility | 6 | Random seeds fixed, environment specified, data accessible, README clear |
| *Presentation (30 marks)* | | |
| Introduction | 5 | Problem clearly framed, justification for using ML, clearly stated research question |
| Methodology | 5 | Dataset introduced; brief model overview; rationale for choosing these models for the task |
| Results & Comparison | 5 | Clear presentation of results with a summary comparison table of all models and their performance |
| Chosen Model & Justification | 5 | Final model clearly identified with a convincing, evidence-based justification |
| Delivery | 5 | Brief, to the point, and fluent; within 10 minutes; all group members contribute |
| Q&A | 5 | Depth of understanding across the whole project; able to justify decisions under questioning |
| **Total** | **50** | |

## 7 Report Format

- Use MS Word or LaTeX for preparing your report.
- Number all sections and subsections.
- Number all pages.
- Ensure the table of contents matches section numbers and page numbers.
- Put captions on all figures, tables, and diagrams.
- Place page breaks appropriately (not immediately after headings or in the middle of a figure/table).
- Cite all work from other sources in the text and list them in your references.
- Most of your report should be written in your own words. Each quotation, table, or diagram from another source must be accompanied by your own discussion of what it shows, why you included it, and why it is relevant.

# 8 Submission

- **Report:** upload into Moodle through the Turnitin Assignment Submission module (under Assessments, "Final Reports") before the due date. The Turnitin similarity score, excluding references, should be less than **10%**.
- **Code and Presentation Slides:** upload into Moodle (under Assessments, "Source Code, Presentation Slides") as one zipped file.
- **Presentation:** delivered in class on a date scheduled separately by the lecturer.

Each group submits only one report, one code archive, and one set of slides. Names of all group members must appear on the first page of the report.

# Policies

## Academic Integrity

Marks will be deducted if you fail to draw upon appropriate professional and academic literature or if the assignment is not written and presented in a professional manner. When choosing sources, check that they have passed some form of quality assurance (books, journal articles, and conference articles are a good choice). If it is difficult to identify the author, year, or title of an article then it has not been subject to quality assurance.

Plagiarism will be reported to the Programme Authorities who may award zero marks or refer the matter to the Discipline Committee, which has powers of suspension/exclusion.

## Late Submission of Assignments

Assignments submitted after the due date and time without having received an extension through Affected Performance Consideration (APC) will be penalised according to the following:

- **10%** of marks deducted if submitted within 24hrs of the deadline.
- **20%** of marks deducted if submitted after 24hrs and up to 48hrs of the deadline.
- **No grade** will be awarded for an assignment that is submitted later than 48hrs after the deadline.

Assignments submitted in more than 48 hours late will not be marked unless Affected Performance Consideration (APC) apply. So, it is better to submit an incomplete assignment on time.

## Affected Performance Consideration

A student, who due to circumstances beyond his or her control, misses a test, final exam or an assignment deadline or considers his or her performance in a test, final exam or an assignment to have been adversely affected, should complete the Affected Performance Consideration (APC) form available from Student Central or online:

https://www.unitec.ac.nz/current-students/study-support/affected-performance-consideration

You must apply within five working days after the due date of your assessment for it to be considered.

## Instructions for Artificial Intelligence Use

For this assessment, you are allowed to use Generative artificial intelligence tools such as ChatGPT, Bard, Grammarly Go etc to:

- Clarify concepts, theories, ideas, etc., discussed in class during your preparation for the assessment.
- Generate preliminary ideas for writing.
- Edit a working draft of the assessment.
- Read and summarise research and supporting evidence for the assessment.

You are NOT allowed to use Generative AI to:

- Generate definitions or writing used in your final submission.
- Produce a paragraph or section of text.
- Produce counter-arguments or refine thinking on your final submission.
- Write any code that is needed for an assessment.

Be aware that using Generative AI, including ChatGPT, for work submitted for assessment, can be considered a type of "third-party assistance" (like getting someone else to write your assignment for you, even a friend or family member). Therefore, submitting AI-generated material as your original work without appropriate acknowledgement may be considered a form of plagiarism, which may constitute academic misconduct.

## Assistance to Other Students

Students themselves can be an excellent resource to assist the learning of fellow students, but there are issues that arise in assessments that relate to the type and amount of assistance given by students to other students. It is important to recognise what types of assistance are beneficial to another's learning and also what types of assistance are unacceptable in an assessment.

**Beneficial Assistance:**

- Study Groups.
- Discussion.
- Sharing reading material.
- Testing another student's programming work using the executable code and giving them the results of that testing.

**Unacceptable Assistance:**

- Working together on one copy of the assessment and submitting it as own work.
- Giving another student your work.
- Copying someone else's work. This includes work done by someone not on the course.
- Changing or correcting another student's work.
- Copying from books, Internet etc. and submitting it as own work.

Do you want to do the best that you can do on this assignment and improve your grades? You could:

- Talk it over with your lecturer.
- Visit Student Success and Achievement for learning advice and support.
- Visit the Centre for Pacific Development and Support.
- Visit the Centre for Maori Development and Support.